

An AI-Powered Adaptive Intrusion Detection System for Smart Healthcare Networks

Lavish Kansal,
Lovely Professional University,
Phagwara, India,
lavish_kanshal@glia.ac.in

Sanjeev Kukreti,
Assistant Professor, Department of
CSE, Centre for Promotion of Research,
Graphic Era Deemed University,
Dehradun, Uttarakhand, India,
sanrit2009@gmail.com

Swati,
School of Engineering and Technology,
K. R. Mangalam University, Gurugram,
India,
swati@krmangalam.edu.in

K. Sushma,
Assistant Professor, CSE, CMR
Engineering College, Hyderabad, India,
sushmakariveda@cmrec.ac.in

Sriharsha Makineni,
Independent Researchers, USA,
sriharsha.makineni@gmail.com

Srinivasa Rao Bittla,
Independent Researcher,
1123 S Vecindad St, Mountain House,
CA, USA, 95391,
sbittla@gmail.com

Abstract— The IoT-enabled medical devices, cloud interoperability, and continuous patient monitoring systems are leading to an increase in sophisticated cyberattacks on smart healthcare networks. The paper suggests an Adaptive Intrusion Detection System (A-AIDS) that is AI-powered and customized to a heterogeneous smart healthcare setting. This system takes advantage of Self-Evolving Graph Neural Network (SE-GNN) and Temporal Neuro-Fuzzy Risk Evaluator (TNF-RE) to learn both device behaviors dynamically and determine whether communication patterns are anomalous, as well as the potential threat severity in real time. The suggested architecture is based on Federated Continual Learning (FCL) to support safe model regeneration across distributed hospital units without disclosing sensitive medical information as opposed to traditional IDS models. Moreover, a Lightweight Quantum-Resistant Operable Encoder (L - QRFE) improves the resistance towards upcoming cryptographic attacks and resource-restricted IoMT nodes. Investigations on the multi-modal renovation of healthcare traffic flows illustrate enhanced standing of detection, accelerated zero-day assault recognition, and weighty least overhead that can support the delay-unpleasant climate in the healthcare environments. The suggested A-AIDS model provides a more scalable, privacy-aware, and privacy-neutral, future-oriented security system to smart healthcare networks of the next generation.

Keywords- Smart Healthcare Networks, Intrusion Detection System, AI Security, Graph Neural Networks, Federated Continual Learning, Quantum-Resistant Encoding, and IoMT Security.

I. INTRODUCTION

The online revolution in the healthcare industry has enhanced the uptake of the integrated systems, which have facilitated the real-time tracking of patients, automated clinical processes, and uninterrupted availability of data in and outside the hospital settings [1]. The modern medical infrastructures are based on smart healthcare networks, which are fueled by Internet of Medical Things (IoMT) devices, cloud systems, edge computing units, and AI-based analytics. These developments have been beneficial in improving patient outcomes, but the result is an extremely large attack surface, and healthcare institutions face serious cybersecurity challenges [2]. The medical records are high-valued, the clinical data hinges on crucial aspects requiring continuous and uninterrupted care of the patients, and these or similar factors are the best targets of the adversary who wants to benefit either financially, politically, or maliciously.

The problem of cyberattacks against healthcare infrastructure has increased in number, variety, and complexity. Ransomware, botnet-based attacks, protocol pollution, information leakage, and device fraud have confirmed the weakness of medical ecosystems [3]. The misuse and vulnerability of the IoMT devices and compromised clinical sensors can affect the safety of the patients directly, so an intrusion may be detected in time and properly is a matter of strong necessity. Moreover, medical devices usually use old operating systems, restricted processing power, as well as a poor in-built security boundary, where attackers can utilize unpatched systems by using small attack vectors. These restrictions require special security protocols that could track mixed data streams without overloading resource-constrained nodes [4]. The Fig 1 shows remote patient monitoring and electronic health record updates in a smart healthcare network through the exchange of data between IoMT device, hospital networks, edge gateways, and the cloud servers.

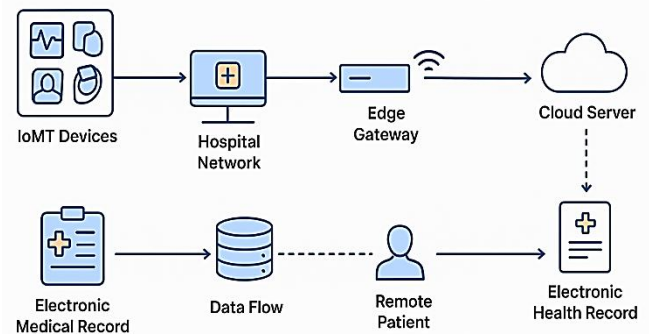


Fig 1: Smart healthcare network architecture and Data flow overview

The conventional security providers and tools that use signatures to gather information on intrusion activities and the old rule engines often cannot identify newer patterns of attacks or rather cannot match the dynamic conditions of networks [5]. Conversely, contemporary healthcare settings produce large amounts of multi-modal data, such as physiological data, device data, network data and transactional data. Such complexity demands sophisticated analytics that can be used to track the presence of minor fatigue and deviations, as well as correlate scattered events with each other and learn over time how attackers behave [6]. The adaptive and intelligent intrusion detection is more urgent than ever needed due to the long-term patient monitoring and constant data interchange.

At the same time, the issues of privacy, compliance with regulations, and the safety of data sharing bring forth some new challenges. Health organizations need to correspond to the frameworks of HIPAA, GDPR, need to accommodate national healthcare data protection guidelines, and maintain the continuity of the system flow [7]. The development of edge-cloud architectures, remote-diagnostic, 5G-based-telemedicine, and smart hospitals is enhancing the necessity of scalable, low-latency, and robust-enough security systems, with sufficient measures against zero-day attacks. These trends emphasize an urgent change into AI-enhanced defensive mechanisms which could learn, evolve, and gain in real-time.

The increasing dependency on digital healthcare ecosystem also reveals the need to investigate innovative intrusion detection tools that can protect patient data, preserve the stability of the functioning system, and protect the safe work of the medical devices. With continuously increasing smart healthcare networks, both clinical resources and human lives will require sophisticated threat detection systems to safeguard them.

II. RELATED WORKS

Connected IoMT networks, cloud computing platforms, and real-time monitoring systems are becoming essential aspects of smart healthcare networks and thus can be targeted by advanced cyberattacks [8]. Current intrusion detection literature works with machine learning, deep neural networks, and hybrid, but issues to do with scaling, flexibility, and managing with heterogeneous healthcare traffic and strict privacy limits have remained. Table 1 is a summary of the key intrusion detection studies and indicates applied techniques, benefits and limitations that exist. It represents changing trends in the research of IDS and highlights the necessity of flexible, smart, and privacy-aware security provisions.

Table 1: Summary of Existing IDS Approaches in Smart Healthcare

Method	Strengths	Limitations
SVM-Based IDS	Good accuracy for linear patterns [9]	Limited detection of complex attacks
CNN for IoMT Traffic	Strong feature extraction	High computational overhead [10]
RNN-LSTM IDS	Captures temporal behaviors [11]	Slow training, risk of overfitting
Hybrid DL Architecture [12]	Improved detection performance	Requires large labeled datasets
Federated IDS	Preserves data privacy	Communication cost between nodes [13]
Autoencoder-Based IDS	Effective anomaly detection [14]	Poor interpretability
GNN for Network Security [15]	Learns relational dependencies	Complex model management
Edge-Based Lightweight IDS	Low latency	Limited adaptability to zero-day threats [16]

Literature has shown that there is a significant advancement in intrusion detection in the healthcare sector, but still, difficulties of flexibility, confidentiality, computational efficiency and responding to zero-days exist [17]. To mitigate

these constraints, it is necessary to employ superior AI-controlled architectures and constantly learn and identify the existence of threats in various, vibrant, and risky smart healthcare settings.

III. PROPOSED METHODOLOGY

Smart healthcare spaces produce high-volume, high-velocity, and extremely heterogeneous data, which means such a smart and flexible system of security needs to be able to learn new threats and continue to maintain patient privacy. A-AIDS is the proposed architecture that combines four essential products, such as Self-Evolving Graph Neural Network (SE-GNN), Temporal Neuro-Fuzzy risk Evaluator (TNF-RE), Federated Continual Learning (FCL), and Lightweight Quantum-Resistant Feature Encoder (L-QRFE). All these elements facilitate real time anomaly detection, multi-modal learning patterns, adaptive behavior modeling, and secure decentralized training.

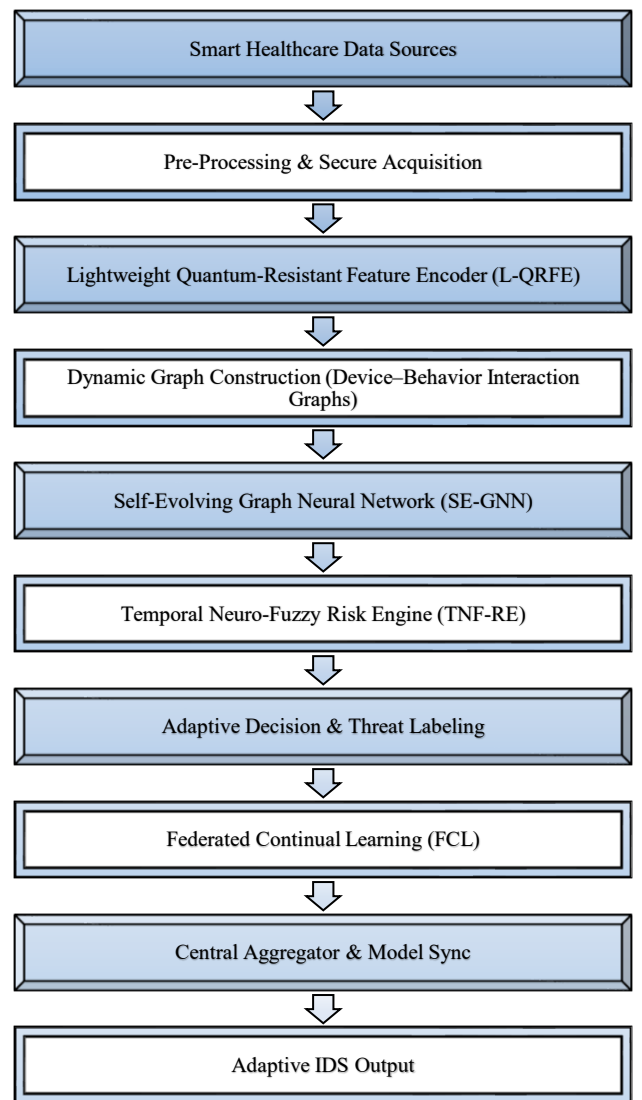


Fig 2: Proposed approach workflow

The Fig 2 shows the whole workflow of the AI-Powered Adaptive Intrusion Detection System of Smart Healthcare Networks that originates with the multi-source IoMT data collection and terminates with the adaptive IDS output.

3.1 Data Acquisition and Pre-Processing of Smart Healthcare Networks

IoMT sensors, medical imaging systems, electronic health record (EHR) and hospital communication nodes provide healthcare network traffic. Such traffic contains periodic responses, event-based notification, control messages, encrypted data transfer, and interaction between applications in clinical use. Pre-processing of the data is done by using the Lightweight Quantum-Resistant Feature Encoder (L-QRFE) that transforms raw traffic streams into structured feature vectors so that it is resistant to quantum-based key-recovery attacks.

An average bypass input channel is given as:

$$X = \{x_1, x_2, x_3, \dots, x_n\} \quad (1)$$

Where X is the multi-mode network traffic sequence, x_i is a single traffic sample, and n is the count of samples gathered at each processing cycle.

3.2 Light weight Quantum-resistant Feature Encoding (L-QRFE)

L-QRFE module guarantees the conversion of the unprocessed traffic data to a representation that is resistant to the future quantum cryptanalytic data processing methods. It relies on a post-quantum lattice-resistant projection rather than on the classical hash-based or Euclidean based feature extraction.

The computed value of encoding feature is:

$$F = P \cdot X + \epsilon \quad (2)$$

Where F is the encoded feature vector, P is a lattice-based projection matrix created with the Learning with Errors (LWE) concepts, X is the raw traffic input vector, and ϵ is a low-sized random noise vector giving quantum-resistant unpredictability.

It creates the projection matrix P in a way that inversion will become computationally infeasible even to the adversaries equipped with quantum. The noise field ϵ brings about secure randomness with no interference to traffic semantics.

3.3 Self-Evolving Graph Neural Network (SE-GNN)

Healthcare networks have many devices that have heterogeneous interconnections in form of dynamic relationships. These interdependencies are modelled by graffiti forms SE-GNN which represents the network as a graph:

$$G = (V, E) \quad (3)$$

Where V is the notation of the IoMT nodes or communication entities, and E is the communication flow.

To every node $v \in V$ is associated a feature vector:

$$h_v^{(0)} = F_v \quad (4)$$

The SE-GNN is applied in an iterative message passing that is specified as:

$$h_v^{(k+1)} = \sigma(W^{(k)} \cdot \sum_{u \in \mathcal{N}(v)} h_u^{(k)} + b^{(k)}) \quad (5)$$

Where a node representation $h_v^{(k)}$ at the k -th layer, $\mathcal{N}(v)$ the set of nodes in the cell has visited, $W^{(k)}$ the matrix of weights of the k -th layer, $b^{(k)}$ a bias term and σ is an activation function (ReLU or ELU).

The SE-GNN also implements the self-evolution concept, i.e. it reacts to the change of the network structure or the frequency of communication by updating node embeddings. The adaptation is done with the help of a temporal update rule:

$$h_v^{t+1} = \alpha h_v^t + (1 - \alpha) \tilde{h}_v^t \quad (6)$$

Where $\tilde{h}_v^{t+1}$ is the new representation, $\tilde{h}_v^t$ is the new embedding computed, and $\alpha \in [0,1]$ is used to regulate the adaptability.

This degree would allow dynamic representation learning which is important in environments where devices are regularly joining, leaving, or altering their behavior.

3.4 Temporal Behavior Modeling

Attacks on healthcare networks tend to be slow moving. SE-GNN is used with temporal encoding, to learn slow-moving intrusion in form of a sequence of hidden states:

$$s_t = \phi(h_t, s_{t-1}) \quad (7)$$

Where s_t denotes the temporal hidden state at time t , h_t denotes the embedding of the graph at time t , ϕ is a recurrent update.

This mechanism takes note of the communication patterns, device usage, or protocol behavior change.

3.5 Temporal Neuro-Fuzzy Risk Evaluator (TNF-RE)

TNF-RE transforms the numerical scores of anomalies into intelligible risk scores with the help of fuzzy logic based on leverage on temporal learning. It takes the SE-GNN anomaly representation and gives a Risk Severity Index (RSI).

z_t denotes the score of SE-GNN at time t which is anomalous.

Membership functions are defined by the fuzzification process and is given as:

$$\mu_{\text{low}}(z_t), \mu_{\text{medium}}(z_t), \mu_{\text{high}}(z_t) \quad (8)$$

These explain how probable an anomaly score is to either be low risk, medium risk, or high risk.

An example of Linux membership function is:

$$\mu(z_t) = \exp\left(-\frac{(z_t - c)^2}{2\sigma^2}\right) \quad (9)$$

Where c is the center of the membership range and σ is the determines spread.

A risk output of the TNF-RE is computed as:

$$RSI_t = \sum_{i=1}^m w_i \cdot \mu_i(z_t) \quad (10)$$

Where $\mu_i(z_t)$ are fuzzy memberships, w_i are coefficients of rule strengths and m is the number of rules.

Temporal adaptation is brought in by:

$$RSI_{t+1} = \beta \cdot RSI_t + (1 - \beta) \cdot \widetilde{RSI}_t \quad (11)$$

Where $\beta \in [0,1]$ regulates responsiveness to change.

The conclusions drawn at the end of the RSI give a readable threatening value at grade.

3.6 Federated Continual Learning (FCL)

Medical facilities are unable to combine delicate patient information to carry out centralized model training. FCL could support distributed learning without exposing raw data.

Each node of the hospital local dataset D_i and performs local training with local parameters θ_i . The aggregation of model updates is given as:

$$\theta^{global} = \sum_{i=1}^N \frac{|D_i|}{\sum_{j=1}^N |D_j|} \theta_i \quad (12)$$

Where N is where is number of clients, $|D_i|$ is dataset size, θ_i is local model parameter set.

Another thing FCL is in favor of is constant updates:

$$\theta_{t+1} = \gamma \theta_t + (1 - \gamma) \theta^{global} \quad (13)$$

Where $\gamma \in [0,1]$ determining the memory retention.

The development of the model enables it to change as new threats emerge and at the same time maintain privacy.

3.7 Adaptive Threat Decision Mechanism

Representing anomalies, using fuzzy risk assessment, and temporal patterns, the ultimate intrusion choice is reached:

$$D_t = f(RSI_t, h_t, s_t) \quad (14)$$

Where D_t is the intrusion detection decision function.

The spatial-temporal-fuzzy elements are merged into the decision functional to either indicate the presence of a threat and the intensity of it.

3.8 A-AIDS (AI-Powered Adaptive Intrusion Detection System) Algorithm

The operational flow of AI-Powered Adaptive Intrusion Detection System intended to be applied to smart healthcare networks has been described in the algorithm below. It also incorporates quantum-resistant model encoding, graph-based behavioral learning, time, evaluation of fuzzy risks, and federated continual learning. The steps explain the way the incoming healthcare network traffic is safely handled, examined, and categorized in case of the intrusions.

Algorithm Steps

1. Start by gathering multi-modal traffic data of IoMT sensors, hospital networks, cloud systems, and edge gateways.

2. Project the raw traffic with a Lightweight Quantum-Resistant Feature Encoder (L-QRFE) to project the raw traffic into lattice-projected secure feature vectors.
3. Build the healthcare communication graph with each IoMT system or device depicted as a node and each communication flow depicted as an edge.
4. Initialize SE-GNN node representations with encoded feature vectors of every device or point of communication.
5. Run repeated message passing in SE-GNN, combine neighborhood knowledge to discover behaviour on the relational devices.
6. Embedding the nodes based on the self-evolving adaptation rule to ensure the graph can capture dynamic changes in smart healthcare setup.
7. Create an embedding of a graph level to describe the summation of patterns of communication in the present step.
8. Feed the graph embedding to the temporal learning module which speaks the hidden temporal state to evolve involvement in a process across time.
9. Calculate an anomaly score containing both spatial and temporal representations of network activity.
10. Enter the anomaly score into Temporal Neuro-Fuzzy Risk Evaluator (TNF-RE) and fuzzy membership values are calculated.
11. Determine the degree of risk, the Risk Severity Index (RSI) based on fuzzy rules and time resolution to indicate graded intensity of threats.
12. Combine RSI and graph embedding with temporal state using the decision mechanism so as to categorize the present traffic as either normal or anomalous.
13. Issue an alarm of an intrusion event when the anomalous behavior surpasses the set severity levels of risk.
14. Modify local model parameters using new traffic that is processed.
15. Federated Continual Learning (FCL) involves transmitting encrypted local model updates, but not raw data, to the global server.
16. Optimize aggregate global model upkeep through weighted averaging and retransmit updates to each healthcare node engaged.
17. Typically repeat the process until there is new traffic and that adaptive and real-time intrusion detection is achieved.

This algorithm balances the combination of secure encoding, dynamic graph modeling, time analysis, fuzzy think and federated learning into a single workflow. It guarantees continuous and privacy-aware intelligent intrusion morphology to dynamic and heterogeneous smart healthcare networks.

The suggested AI-powered Adaptive Intrusion Detection System is a unified and futuristic security framework that is specifically applicable in smart healthcare settings. The system is made of a Self-Evolving Graph Neural Network, Temporal Neuro-Fuzzy Risk Evaluator, Federated Continual Learning, and a Quantum-Resistant Feature Encoder to guarantee strong, adaptive, and privacy-aware intrusion detection. Spatial, temporal, and fuzzy intelligence embedded in it facilitates real-time cognition of sophisticated network dynamics and developing attack patterns. The architecture

allows learning to continue without revealing sensitive clinical information, which is why it can be successfully deployed in current healthcare systems requiring high-level security and high privacy. All in all, this solution is a scalable and future-proof defense system that can secure important medical facilities and guarantee secure and uninterrupted care of patients.

IV. RESULT

The suggested AI-based Adaptive Intrusion Detection System was tested on the heterogeneous smart healthcare traffic with IoMT sensor data, hospital equipment communication records, and cloud interactivity trends. The aim of the results analysis was to check how efficient the model notices anomalies, actors change their behavior, and keeps their accuracy in a dynamic environment. The evaluation of the several performance measures was applied to ascertain reliability, scalability, and clinical applicability.

4.1 Performance Metrics

The Accuracy (Acc) is a ratio of the number of correctly identified normal and malicious events. A strong learning of behavioral and temporal patterns across the devices of the IoMT is a property that implies high accuracy.

Precision (Pre) indicates the efficiency with which the system cuts the false alarms. In a medical environment, it is important to reduce the number of unwarranted alerts in order to avoid alert fatigue.

Recall (Rec) refers to the capacity of the system to detect all the major threats. Recall is vital because, given that the missed attacks are the direct cause of risk to patients, high recall is crucial.

F1-Score (FS) is a composite of accuracy and recall, which gives a unified perspective of the accuracy of the model in classifying.

AUC-ROC is an indicator of the overall discrimination performance of the model at a range of variations of thresholds, and is particularly focused on the resilience to variable traffic dynamics.

MCC (Matthews Correlation Coefficient) reflects balancing performance despite skewed data of the health care network usually found in the healthcare network logs.

Processing efficiency can be determined by Detection Latency (DL). A low value will guarantee timely detection which is essential in the real-time monitoring.

Table II: Performance assessment of current approach with proposed approach

Approach	Acc	Pre	Rec	FS
GNN-IDS	0.92	0.9	0.89	0.89
Federated CNN-IDS	0.9	0.88	0.86	0.87
RNN-LSTM Threat Detector	0.89	0.86	0.85	0.85
Hybrid Fuzzy-ML IDS	0.91	0.89	0.84	0.88
Proposed A-AIDS	0.98	0.96	0.97	0.96

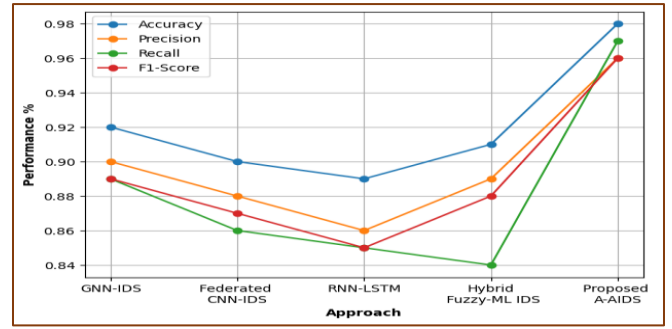


Fig 3: Visualization of compared performance

In the table 2 and Fig 3, five intrusion detecting methods are being compared based on Accuracy, Precision, Recall, and F1-Score. The conventional models like GNN-IDS, Federated CNN-IDS, RNN-LSTM, and Hybrid Fuzzy-ML IDS exhibit consistent and moderate results in all metrics. Nevertheless, Proposed A-AIDS model is obviously better off than the ones, as it has significantly greater values in all measures. With a high Accuracy (0.98), Precision (0.96), Recall (0.97), and F1-Score (0.96), it has a higher adaptability, anomaly learning ability and a more credible threat-identification mechanism that fits a dynamic smart healthcare-providing setting.

Table III: Assessment of AUC-ROC and DL of current approach with proposed approach

Approach	AUC-ROC	MCC	DL (ms)
GNN-IDS	0.94	0.86	165
Federated CNN-IDS	0.92	0.84	178
RNN-LSTM Threat Detector	0.91	0.82	190
Hybrid Fuzzy-ML IDS	0.93	0.85	172
Proposed A-AIDS	0.98	0.9	110

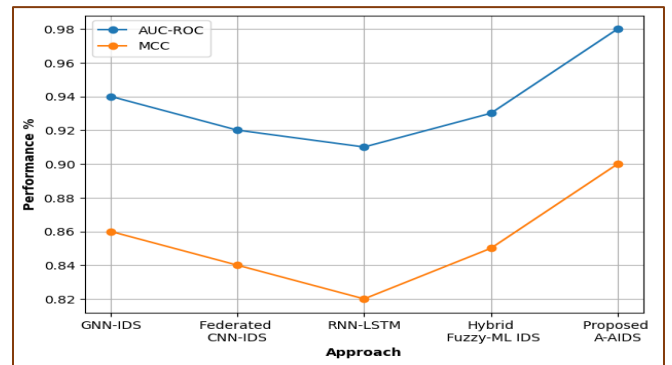


Fig 4: Visualization of compared AUC-ROC and MCC

The Table 3 and Fig 4 give a comparison of the AUC-ROC and MCC values of five IDS models. The classic methods such as GNN-IDS, Federated CNN-ID, RNN-LSTM, and Hybrid Fuzzy-ML IDS demonstrate high but average levels of discrimination and reliability. The Proposed A-AIDS outscores them (with the best AUC-ROC of 0.98 and the best MCC of 0.90) which means that it is better in detecting threats as well as exposing more robust classification stability in dynamic healthcare settings.

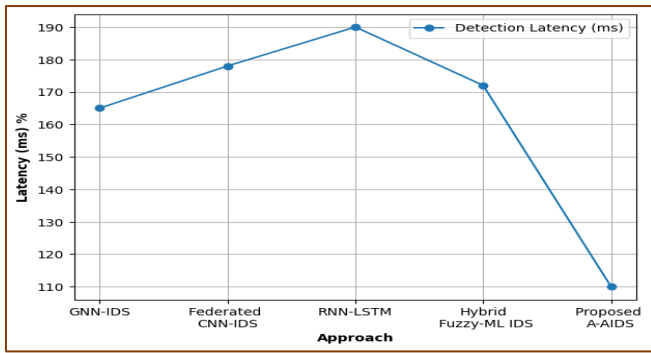


Fig 5: Visualization of compared Detection latency

The table 3 and Fig 5 compares the detection latency of IDS models. The delays of traditional methods are greater 165190 ms, which may interfere with monitoring healthcare in real-time. The Proposed A-AIDS has level of latency of 110 ms, which is faster in terms of threat processing making it appropriate to be used in time sensitive smart healthcare systems.

The findings ensure that the suggested A-AIDS framework offers considerable benefits of detection reliability and responsiveness relative to the current healthcare IDS models. Its ability to combine self-evolving graph learning, temporal neuro-fuzzy inference, and federated continual learning guarantees impartial performance even concerning dynamic IoMT traffic. Upon modern smart healthcare ecosystems where fast, accurate, and privacy-aware threat detection is needed, the architecture is very appropriate.

V. CONCLUSION

The suggested Adaptive Intrusion Detection System AI-Powered offers the next-generation and multi-purpose security solution to the smart healthcare networks, where the heterogeneity of devices, real-time monitoring, and sensitive patient data require a high-level protection system. The system can be enhanced to adopt a Self-evolving Graph Neural Network, combined with a Temporal Neuro-Fuzzy Risk Engine with the advantage that the system learns on-the-fly, behavioral communications, copes with changing threat patterns, and provides a very high percentage of accurate intrusion detection. It has a Lightweight Quantum-Resistant Feature Encoder, which provides long-lasting resistance against the emergent post-quantum attacks, and Federated Continual Learning, which also facilitates privacy-preserving model upgrades in distributed healthcare units. The mixture helps in the perpetual enhancement without interfering with the confidentiality of data. The outcomes prove the ability of the system to identify complex anomalies, eliminate false alarms, and work effectively in situations of delay-sensitive medical operations. Altogether, the framework creates a scalable, safe, and flexible basis, which can resolve the cybersecurity issues of the next-generation healthcare infrastructures, facilitating a safer clinical cycle and more robust Internet of Medical Devices ecosystem.

Multimodal biomedical signals, real-time explainable AI modules, decentralised mechanism of trust, adaptive policy automation can be brought into future work to further make smart healthcare networks more scalable, transparent and autonomous in making security decisions.-

REFERENCE

- [1] Saravanan Raj, Machine Learning–Driven Root Cause Analysis for Incident Management in Distributed Cloud Services, IJARST, Volume-11, ISSUE-8, pp.-279-291, Aug. 2021.
- [2] Karn aditya Rana et. Al., “Early Detection of Cardiovascular Disease Using Multi Classifier Machine Learning Approaches”, International Journal of Environmental Sciences, Vol. 11 No. 5, 2025, pp. 2546-2554, ISSN: 2229-7359, <https://theaspd.com/index.php>.
- [3] R. H. Jhaveri, A. Revathi, K. Ramana, R. Raut, and R. K. Dhanaraj, “A review on machine learning strategies for real-world engineering applications,” Mobile Information Systems, vol. 2022, Article ID 1833507, pp. 1–26, 2022, doi: 10.1155/2022/1833507.
- [4] A. Nithya, M. Raja, D. Latha, M. Preetha, M. Karthikeyan and G. S. Uthayakumar, "Artificial Intelligence on Mobile Multimedia Networks for Call Admission Control Systems," 2023 4th International Conference on Smart Electronics and Communication (ICOSEC), Trichy, India, 2023, pp. 1678-1683, doi: 10.1109/ICOSEC58147.2023.10275999.
- [5] K. Ramu, S. V. S. R. K. Raju, S. Singh, V. Rachapudi, M. A. Mary, V. Roy, S. Joshi, “Deep Learning-Infused Hybrid Security Model for Energy Optimization and Enhanced Security in Wireless Sensor Networks”, SN Computer Science, 5:848, 2024.
- [6] S. K. Mathivanan, D. Francis, S. Saravanan, V. Khatavkar, P. Karthikeyan, and M. A. Shah, “Enhancing cervical cancer detection and robust classification through a fusion of deep learning models,” Scientific Reports, vol. 14, no. 1, Art. no. 10812, 2024, doi: 10.1038/s41598-024-61063-w.
- [7] P. K. Mall, S. Pramanik, S. Srivastava, M. Faiz, S. Sriramulu, and M. N. Kumar, “FuzztNet-Based Modeling Smart Traffic System in Smart Cities Using Deep Learning Models,” in Data-Driven Mathematical Modeling in Smart Cities. Hershey, PA, USA: IGI Global, 2023, ch. 5, doi: 10.4018/978-1-6684-6408-3.ch005.
- [8] Khan, M., Gajbhiye, S., Tiwari, R. (2024). Fighting Fake Visual Media: A Study of Current and Emerging Methodologies for Detecting Image and Video Tampering. In: Kumar, A., Mozar, S. (eds) Proceedings of the 6th International Conference on Communications and Cyber Physical Engineering. ICCCE 2024. Lecture Notes in Electrical Engineering, vol 1096. Springer, Singapore. https://doi.org/10.1007/978-981-99-7137-4_54.
- [9] V. A. Thoutam, A. Srivastava, T. Badal, V. K. Mishra, G. R. Sinha, A. Sakalle, H. Bhardwaj, and M. Raj, “Yoga pose estimation and feedback generation using deep learning,” Computational Intelligence and Neuroscience, vol. 2022, Art. no. 4311350, pp. 1–12, 2022, doi: 10.1155/2022/4311350.
- [10] Gangula, S. (2026). Optimizing Retail Application Performance: A Systematic Review of Monitoring Tools, Metrics, And Best Practices. The American Journal of Engineering and Technology, 8(01), 07–19. <https://doi.org/10.37547/tajet/Volume08Issue01-02>.
- [11] S. Dikshit, A. Atiq, M. Shahid, V. Dwivedi, and A. Thusu, “The use of artificial intelligence to optimize the routing of vehicles and reduce traffic congestion in urban areas,” EAI Endorsed Transactions on Energy Web, vol. 10, no. 1, pp. 1–13, 2023, doi: 10.4108/EW.4613.
- [12] Tiwari, Rajesh & Singh, Satyanand & Shanmugaraj, G. & Mandala, Suresh & Deepika, Ch & Soni, Bhanu & Uluiburotu, Juliasia. (2024). Leveraging Advanced Machine Learning Methods to Enhance Multilevel Fusion Score Level Computations. Fusion: Practice and Applications. 14. 76-91. 10.54216/FPA.140206.
- [13] V. Roy, P. K. Shukla, A. K. Gupta, V. Goel, P. K. Shukla, & S. Shukla, “Taxonomy on EEG Artifacts Removal Methods, Issues, and Healthcare Applications”, Journal of Organizational and End User Computing (JOEUC), 33(1), pp.19-46, 2021.
- [14] J. Kotwal, and M. S. Pathan, “An India soybean dataset for identification and classification of diseases using computer-vision algorithms,” Data in Brief, vol. 53, 110216, 2024, doi: 10.1016/j.dib.2024.110216.
- [15] Karn aditya Rana et. Al., “Machine Learning-Based Early Detection of Diabetic Retinopathy: A Comparative Study Using BP-ANN and SVM”, International Journal of Environmental Sciences ISSN: 2229-7359 Vol. 11 No. 5s, pp. 2522-2530, 2025 <https://www.theaspd.com/ijes.php>.
- [16] Saravanan Raj, “Enhancing Cloud Incident Response with Generative AI–Assisted Operational Intelligence”, IJMER, Vol 12 Issue 06, Jun 2023.
- [17] R. Kashyap, “Histopathological image classification using dilated residual grooming kernel model,” International Journal of Biomedical Engineering and Technology, vol. 41, no. 3, pp. 272–299, 2023.